

Avnish Pal

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Experience

Software Engineer (Contractor)

Jul 2024 – Present

Autodesk

Singapore

- Built a JWT-based offline authorization cache for the client-side licensing service, enabling compliance checks in Autodesk products without internet access.
- Migrating a Fargate-based Java microservice that every Autodesk product depends on to a new AWS environment, replacing manual deployment with Terraform and YAML-driven infrastructure as code.
- Cut third-party library update time from 3 days to 1 day by building a Python multi-agent system (Jira/Confluence MCP, Cursor CLI agents) that automates update PRs.
- Served as Release Manager for 2 Licensing releases shipped to about 1M users, coordinating 6 feature teams.

Software Quality Engineer

Jul 2021 – Jun 2024

Red Hat

Pune, India

- Cut certification test time from 120 to 40 minutes, with no manual input, by automating OpenShift Operator, Helm chart and container certification pipelines with Tekton and Go.
- Built a Go CLI that provisions OpenShift Pipelines and the operator under test on a target cluster, then runs the certification pipeline and its validation steps in-container. Inner-sourced across teams for certification testing.
- Caught inconsistencies between product metadata, the Red Hat Ecosystem Catalog, API responses and container registry manifests by adding automated validation to the pipeline that certifies 953 partner products.
- Built a Grafana dashboard and metrics service tracking certification pipeline performance.

Software Engineer, Machine Learning (Part-Time)

Sep 2019 – Jun 2021

Taiyo AI

Delhi, India

- Built a Python ML platform that produced investment insights from large financial datasets for stock analysis and portfolio optimization.
- Built a scraper that collected economic indicators from multiple sources for supply-chain risk assessment.
- Built REST APIs serving real-time commodity market data.

Projects

Chat with Document | Python, Streamlit, LangChain, Ollama, Generative AI

- Developed a web application enabling users to upload documents and interact with them using natural language queries through integration with Ollama's self-hosted large language models.

Command Line Tool for Journaling | Go, Cobra, CLI

- CLI for journaling daily accomplishments and generating periodic reports (daily, weekly, monthly, bi-monthly), providing users with an organized method for tracking progress and reflecting on achievements.

OpenCourser | Flutter, Dart, Firebase

- Open-source mobile app where users submit free courses for a skill and rank them by community upvotes.

Open Source Contributions

Hamilton: Containerized examples for Hamilton, a micro-framework for creating dataflows.

Flaskerizer: Minimized complexity and improved performance of hyperlink parsing by 7x.

Education

Bachelor of Technology (Information Technology)

2017-2021

Guru Gobind Singh Indraprastha University

Delhi, India

Skills

Programming Languages: Go, Python, C++, JavaScript, SQL, Bash

Libraries & Frameworks: LangChain, scikit-learn, pandas, Django, Flask, Apache Airflow

Cloud & Infrastructure: Kubernetes, OpenShift, Podman, Docker, Jenkins

Database & Tools: Git, GitHub, GitLab, PostgreSQL, MySQL, SQLite, MongoDB, Grafana

Platforms: GNU/Linux, AWS, GCP, Terraform, Ansible

Languages: English (Fluent), German (Beginner)